

Nutrition iN elderly



BY

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Proper nutrition is important for all of us. It may be even more important for our elderly loved ones. Malnutrition is more common in elderly persons than in younger adults .

 physiological Changes during aging and affect nutritional intake :

- 1- Loss of Teeth
- 2- Decreased Neuromuscular coordination
- 3- Impaired hearing and failing vision
- 4- Diminished sense of taste and smell
- 5- Anorexia
- 6- Physical Discomfort
- 7- Change in Body Composition
- 8- Change in gastro-intestinal tract
- 9- Change in cardiovascular system & Change in respiratory function
- 10- Change in renal function & Change in skeletal tissue
- 11- Socio Psychological: Changes Food habit
Economic aspects , Loneliness .
- 12- Lack of nutritional knowledge
- 13- Depression & Anxiety
- 14- Loss of self-esteem also Loss of independence .

15- Antidepressants, antihypertensives, and bronchodilators can contribute to xerostomia .

16- Difficulty cooking due to disabilities.

17- Lack of access to transportation to buy food.

18- Medications : It is important to follow recommendations regarding medications and eating ; Some medications need to be consumed with food, some need to be consumed on an empty stomach

19- Medications May Affect Nutrient Needs

20- Older adults should follow their doctor's advice about eating, especially if they have an illness that requires changes in what or how they eat, or if they are taking medications. ;

- Some medications can interact with certain foods, or can affect appetite
- Some foods can interact with medications
- Some medications can cause water loss
- Some medications can cause depression

Essential nutrients Requirements for elderly :



Water :

► Water, Water, Water

Make sure the person you are caring for has plenty of water to avoid dehydration, which can lead to appetite suppression.



Image: <http://maulimyth.blogspot.in/2012/10/water-101.html>

While 70% of the body is water in the adult period, this amount decreases to 50% in elderly. • Loss of 15% of body water cause loss of life. • [it Helps in](#): — Digestion, absorption, transportation of food eaten. — Operation of cells, tissues and organs — Monitoring of body heat — Sickness of the joints — Removal of harmful substances from the body.

Dehydration Signs

0-1%	Thirst
2-5%	Dry Mouth, Flushed Skin, Fatigue, Headache
6%	Increased Body Temperature, Breathing, Pulse Rate
8%	Dizziness, Increased Weakness, Labored Breathing
10%	Muscle Spasms, Swollen Tongue, Delirium
11%	Poor Blood Circulation, Failing Kidney Function



Fibre

- Prevention of overweight problems .
- Protection from intestinal cancer.
- Lowering blood cholesterol.
- Increases activities of intestine and prevention of constipation.
- Regulation of blood sugar and protection from the diabetes.
- Sources: Legumes, grains, vegetables and fruits
- 25-30g of fibre consumption per day is sufficient.



Energy :

- Energy requirement reduces
- Basal metabolic rate decreases (15-20 percent) due to reduced muscle mass and other metabolically active tissue mass
- . • Reduced physical activity

- Increase in fatty tissue

Sedentary Moderate Male 1883 kcal -2216 kcal

Female 1706 kcal- 2007 kcal



Carbohydrate



- Requirement reduces.
- Impaired glucose intolerance can lead to hypoglycemia, hyperglycemia, and type II diabetes mellitus.
- Insulin sensitivity can be enhanced by balance energy intake, weight management and regular physical activity.
- 50 percent energy should derive from carbohydrate.



Protein:

- Decreased skeletal tissue mass.
- Decrease in store of protein is inadequate to meet the need of protein synthesis.
- Intake of 1.0gm/kg is safe during old age.
- Protein rich food with milk and curd should be included.



- Due to decrease appetite and poor digestion, old people consume less protein which may lead to edema, anemia, and low resistance to infection



Lipid :

- Dementia and CVD may share risk factors like high intake of dietary total fat.
- Emphasis should be placed on reducing the intake of saturated fat and choosing mono saturated or poly saturated fat sources.
- Sufficient intake of ω -3 fatty acids helps in visual acuity, hair loss, tissue inflammation, improper digestion, poor kidney function and mental depression.



Minerals

- Calcium: 800mg/day To compensate age related bone loss, to improve calcium balance and to decrease prevalence of fracture. Ca absorption efficiency decreases, vitamin D level decreases so need more Ca. Total food consumption decrease so Ca supplements needed.
- Iron: 30mg/day Deficiency is seen in elderly due to inadequate iron intake, blood loss due to chronic disease or reduced non-heam iron absorption.

Vitamin C deficiency also reduce iron absorption. Mild anemia affect health due to less efficient circulation of blood.

- Magnesium : is needed for bone and tooth formation, nerve activity, and metabolism of carbohydrates, protein and fat , DRI for magnesium for adults 51 years of age and older is 420 milligrams for males and 320 milligrams for females per day ,. The UL for magnesium is 350 milligrams from supplements and medications.
- Leading food sources of magnesium : are milk, yeast bread, coffee, ready-to-eat cereal, beef and potatoes.
- Zinc : Some features like delayed wound healing, decreased taste sensitivity and anorexia are associated with zinc deficiency. and immune response, and increases risk of dermatitis , DRI for zinc for adults 51 years of age and older is 8 milligrams for females and 11 milligrams for males per day ,The UL for zinc is 40 milligrams per day, food sources of zinc are beef, ready- to-eat cereals, milk and

poultry , But healthy elderly don't show zinc deficiency.

- **Vitamins** : • Elderly are at risk of Vitamin D deficiency due to decreased exposure to sunlight or decrease in renal mass. • Dietary supplements with calcium and vitamin D improves bone density and prevent fractures. • People with Parkinson disease have low vitamin D levels.

Vitamin C : • Stress, smoking, and medication can increase vitamin c requirement. • The antioxidant vitamins such as vitamin E, carotinoids and vitamin c enhances health of elderly. • Vitamin C may be protective against cataract at an intake level of b/w 150-250 mg/day. • Vitamin E is potent nutrient for reducing decline in cellular immunity that occur in elderly.

Other Vitamins • Changes in immune system can be overcome by taking 200 mg of vitamin E. • Requirement of vitamin B6 increased due to atrophic gastritis, interferes with absorption. • Alcoholic and liver dysfunction are additional risk factor for deficiency of

vitamin B6. • Vitamin B6 have significant role in immune system.

Folic Acid : • Alcoholism is a risk factor for folate deficiency. • Severe deficiency of folic acid may result anemia and elevated serum homo-cystiene level which is a risk for cardiac diseases. • consumption of folate rich food is needed. • Causes for vitamin B12 deficiency are atrophic and bacterial overgrowth, which decreases absorption and leads to pernicious anemia.

Dietary tips :

1. Keep it small Instead of three large meals a day, which can look overwhelming to someone in poor health, better serve six small meals a day.
2. Bulk up on the amount of calories per meal
For instance, you can add protein powder mix to shakes or drinks to increase calories.
3. (Soft is better) Serve soft foods such as pudding, ice cream or fruit smoothies because they can be tasty and easy to digest.
4. Counseling and education: involving dietary modifications, activity recommendations,

medication management, self-monitoring, prognosis, coping skills, social support, caregiver stress, and spiritual needs are critical components maintaining a healthy lifestyle.

5. Make it tasty Don't serve bland or sour tasting foods.
6. Put the power in their hands When possible, give the person you are caring for the decision-making power to decide what they would like to eat; it helps them to feel in control.
7. Make it pretty Present appetizing looking meals by accenting the plate with a garnish (i.e. strawberry or melon). Also, make the dining experience pleasant for the person you are caring for by playing soft music or talking to them about the day's events while they are eating to take their minds off not feeling well.
8. Write it down Keep a food diary about the person you are caring for and include: what food they have problems or complications digesting and their daily food menus, and review it with their doctor or dietician for feedback. They may be experiencing digestive

problems or irritable bowel syndrome due to their menu.

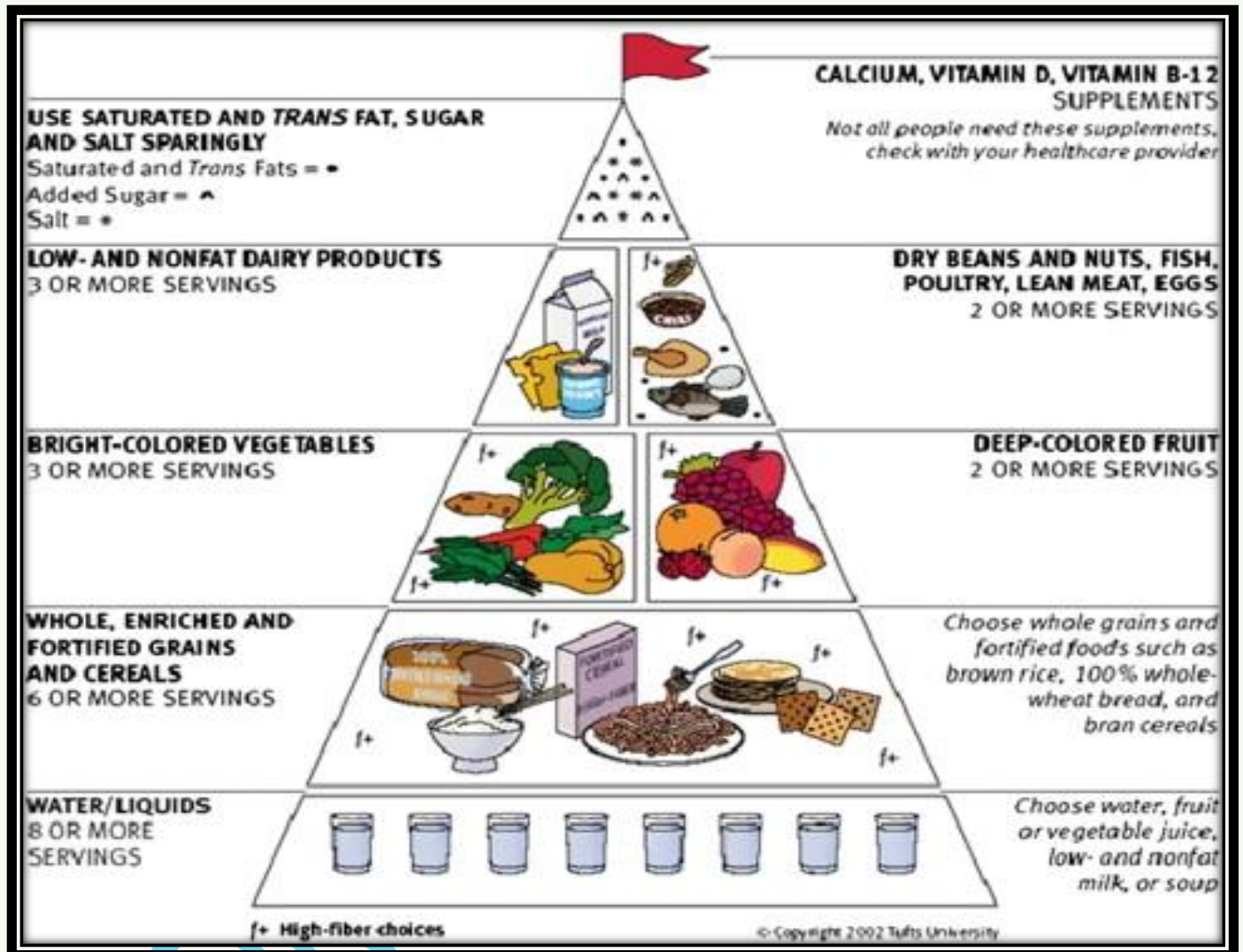
9. Work it out Try and get them moving to work up an appetite. If overall exercise such as walking isn't possible, have them fold the laundry or peel vegetable.



NUTRITION RELATED PROBLEMS AMONG ELDERLY :

- Obesity • Under nutrition and malnutrition • Osteoporosis • Cardiovascular diseases • Diabetes • Cancer
- ▶ Dietary fat seems to be associated with **cancer of the colon, pancreas and prostate.**
- ▶ Blood pressure, blood lipids and glucose tolerance are significantly affected by dietary factors.
- ▶ Malnutrition (particularly vitamin B12, B6, and folate), are associated with cognitive impairment in the elderly.
- ▶ The decreased ability to prepare a meal, has been cited as one of the earliest signs of mild cognitive impairment (MCI)
- ▶ For persons with moderate to severe Alzheimer disease, forgetting to eat, inability to access food, and apraxia with utensils may further impair oral intake.

Food pyramid for an elderly



Screening and assessment of nutritional status in the elderly:

Clinical Observations

- General appearance
- Normal, alert and responsive
- General vitality
- Endurance, energy, good sleep habits
- Weight
- Look at height, age and body build(BMI)
- Muscles Well developed, firm , good tone,
- some fat under skin
- Posture
- Erect
- Skeleton
- No malformations
- Bowlegs, knock-knees, chest deformity
- Legs, feet
- No edema, tenderness or tingling
- Good color
- Nails Firm, pink, no spoon shape or ridges
- Skin :Smooth, moist, good color
- Hair : Shiny, healthy scalp
- Face and neck : Uniform color, healthy, no swelling

- Eyes : Bright, clear, shiny
- Mouth : Lips smooth, moist, good color
- Gums pink, healthy, Tongue pink to red, smooth
- Nervous system control
- Good attention span,
- not irritable or restless,
- normal reflexes
- Gastrointestinal function
- Good appetite and digestion
- Regular elimination pattern
- No palpable organs or masses
- Cardiovascular function : Normal rate and rhythm, blood pressure
- ▶ The most widely used is the Mini-Nutritional Assessment (MNA). Risk of malnutrition (score 17–23.5), and malnourishment (score < 17).
- ▶ Other screening tools is the body mass index (BMI = weight in kilograms (kg)/height in meters (m)²).
- ▶ Another screening tool is the SCALES assessment : Sadness (depression); Cholesterol levels; Albumin (serum levels <4 g/L); Loss of weight; Eating

problems (cognitive and/or physical determinants); and Shopping problems or inability to prepare a meal.

- ▶ A problem with three or more of these areas reflects a high risk for malnutrition.
- ▶ Another tool is the “Determine Checklist” which can be a very useful tool in the community setting.
- ▶ In the nursing home setting, a validated tool is the amount of food left on a resident's plate

Nutrient Supplements: When ? • May be useful with those who: • Lack appetite resulting from illness, loss of taste or smell, or depression • Have diseases in GI tract • Have a poor diet due to food insecurity, loss of function, or disinterest • Avoid specific food groups • Take medication or other substances that affect absorption or metabolism